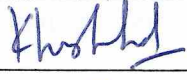
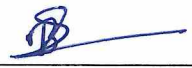


PART-B

Name of the Student: KHUSHHAL BANSAL
Signature of the Student:  Roll No.: 230123031
Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	6	7	5	2	20

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

$$S = \begin{bmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{bmatrix} \quad S_{yX} \text{ is } q \times 1$$

Since r_1^2 is highest eigen value of

$$\sum_{11}^{-1/2} \sum_{12} \sum_{22}^{-1} \sum_{21} \sum_{11}^{-1/2} \rightarrow \text{Since } 1 \times 1 \text{ matrix so equal } r_1^2 \text{ only}$$

$$\sum_{11} = (S_{yy})_{1 \times 1}$$

$$\sum_{12} = (S_{yX}^T)_{1 \times q}$$

$$\sum_{22} = (S_{XX})_{q \times q}$$

$$\sum_{21} = (S_{yX})_{q \times 1}$$

$$\text{Instead of } X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$X' = \begin{bmatrix} x_{11} - \bar{x}_1 \\ \vdots \\ x_{n1} - \bar{x}_1 \end{bmatrix}$$

$$r_1^2 = S_{yy}^{-1/2} S_{yX}^T S_{XX}^{-1} S_{yX} S_{yy}^{-1/2} \quad \text{--- (1) ---}$$

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i2} & \dots & x_{in} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix}$$

Continued on last

$$A = \begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \dots & x_{1n} - \bar{x}_n \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \dots & x_{2n} - \bar{x}_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{i1} - \bar{x}_1 & x_{i2} - \bar{x}_2 & \dots & x_{in} - \bar{x}_n \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \bar{x}_1 & x_{n2} - \bar{x}_2 & \dots & x_{nn} - \bar{x}_n \end{bmatrix}$$

where $\bar{x}_i = \frac{\sum_{j=1}^n x_{ij}}{n}$

$$B = \begin{bmatrix} y_1 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{bmatrix}$$

$$\bar{y} = \frac{\sum y_i}{n}$$

on regression of B on A $\rightarrow B = AB + E$

$$R^2 = \frac{SS_{\text{Reg}}}{SS_{\text{TT}}} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

which

$$R^2 = \frac{SS_{\text{Reg}}}{SS_{\text{TT}}} = \frac{\sum (\hat{b}_i - \bar{b})^2}{\sum (b_i - \bar{b})^2} \quad \text{--- (1)}$$

$$B = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

$$\bar{b} = \frac{\sum b_i}{n} = 0$$

$$R^2 = \frac{\sum (\hat{b}_i - \bar{b})^2}{\sum (y_i - \bar{y})^2} = \frac{\sum \hat{b}_i^2}{SS_{yy}(n-1)}$$

$\hat{b}_i$ is fitted value of b on i^{th} observation (i^{th} row of A)

or $\bar{b} = 0, b_i = y_i - \bar{y}$

Continued on last

Page - 9

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

a) Highest 2 eigenvalues are 4.2, 2.1

$m=2$

$$\tilde{\mathbf{L}} = \begin{bmatrix} \sqrt{\lambda_3} \mathbf{e}_3 & \sqrt{\lambda_2} \mathbf{e}_2 \end{bmatrix}$$

$\downarrow$ $\downarrow$
 1st column 2nd column
 $\tilde{\mathbf{L}}$ $\tilde{\mathbf{L}}$

02

b) $\text{Var}(X_i) = \mathbf{L} \mathbf{L}' + \Psi$

$$\text{Var}(X_1) = \begin{bmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \\ l_{31} & l_{32} \\ l_{41} & l_{42} \end{bmatrix} \begin{bmatrix} \\ \end{bmatrix} + \Psi_1$$

$$\text{Var}(X_1) = l_{11}^2 + l_{12}^2 + \Psi_1$$

$$\begin{aligned} \text{Communality} &= l_{11}^2 + l_{12}^2 = (0.8)^2 + (0.4)^2 \\ &= 0.64 + 0.16 \\ &= 0.8 \end{aligned}$$

01

1) let $e_3 = \begin{bmatrix} e_{31} \\ e_{32} \\ e_{33} \\ e_{34} \end{bmatrix}$ $e_2 = \begin{bmatrix} e_{21} \\ e_{22} \\ e_{23} \\ e_{24} \end{bmatrix}$

~~$h_i^2 = e_{3i}^2$~~

$$L = \begin{bmatrix} \sqrt{\lambda_3} e_{31} & \sqrt{\lambda_2} e_{21} \\ \sqrt{\lambda_3} e_{32} & \sqrt{\lambda_2} e_{22} \\ \sqrt{\lambda_3} e_{33} & \sqrt{\lambda_2} e_{23} \\ \sqrt{\lambda_3} e_{34} & \sqrt{\lambda_2} e_{24} \end{bmatrix} \Rightarrow LL^T = \begin{bmatrix} \sqrt{\lambda_3} e_{31} & \sqrt{\lambda_2} e_{21} \\ \sqrt{\lambda_3} e_{32} & \sqrt{\lambda_2} e_{22} \\ \sqrt{\lambda_3} e_{33} & \sqrt{\lambda_2} e_{23} \\ \sqrt{\lambda_3} e_{34} & \sqrt{\lambda_2} e_{24} \end{bmatrix} \begin{bmatrix} \sqrt{\lambda_3} e_{31} & \sqrt{\lambda_3} e_{32} & \sqrt{\lambda_3} e_{33} & \sqrt{\lambda_3} e_{34} \\ \sqrt{\lambda_2} e_{21} & \sqrt{\lambda_2} e_{22} & \sqrt{\lambda_2} e_{23} & \sqrt{\lambda_2} e_{24} \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_3 e_{31}^2 + \lambda_2 e_{21}^2 & \dots \\ \dots & \dots \end{bmatrix}$$

$$h_i^2 = (\sqrt{\lambda_3} e_{3i})^2 + (\sqrt{\lambda_2} e_{2i})^2 \rightarrow \text{Squared Sum of } i^{\text{th}} \text{ row of } L$$

or i^{th} diagonal element of LL^T

$$\sum_{i=1}^4 (h_i)^2 = \sum_{i=1}^4 (\sqrt{\lambda_3} e_{3i})^2 + \sum_{i=1}^4 (\sqrt{\lambda_2} e_{2i})^2$$

$$= \sum_{i=1}^4 \lambda_3 e_{3i}^2 + \sum_{i=1}^4 \lambda_2 e_{2i}^2$$

$$= \lambda_3 (e_3^T e_3) + \lambda_2 (e_2^T e_2)$$

$$= \lambda_3 + \lambda_2$$

$$\begin{aligned} & \text{as } e_{31}^2 + e_{32}^2 \\ & + e_{33}^2 + e_{34}^2 = e_3^T e_3 \end{aligned}$$

$$e_3^T e_3 = e_2^T e_2 = 1$$

(normalized vectors)

λ_2, λ_3 are Highest two eigen values
Hence proved.

04

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

From CLT

From CLT

$$\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

From delta theorem let g be a differentiable function

$$\sqrt{n} \left(g(\bar{x}_n) - g(u) \right) \xrightarrow{d} N \left(0, (g'(u))^2 u^2 \right)$$

For ~~the~~ asymptotic ~~the~~ variance to be independent of n

independ.

$$(g'(u))^2 u^2 = C \quad \forall u \neq 0 \quad (C \text{ is constant})$$

if $u=0$

$$(g'(u))^2 = \frac{c}{u^2}$$

$$|g'(u)| = \frac{c}{|u|}$$

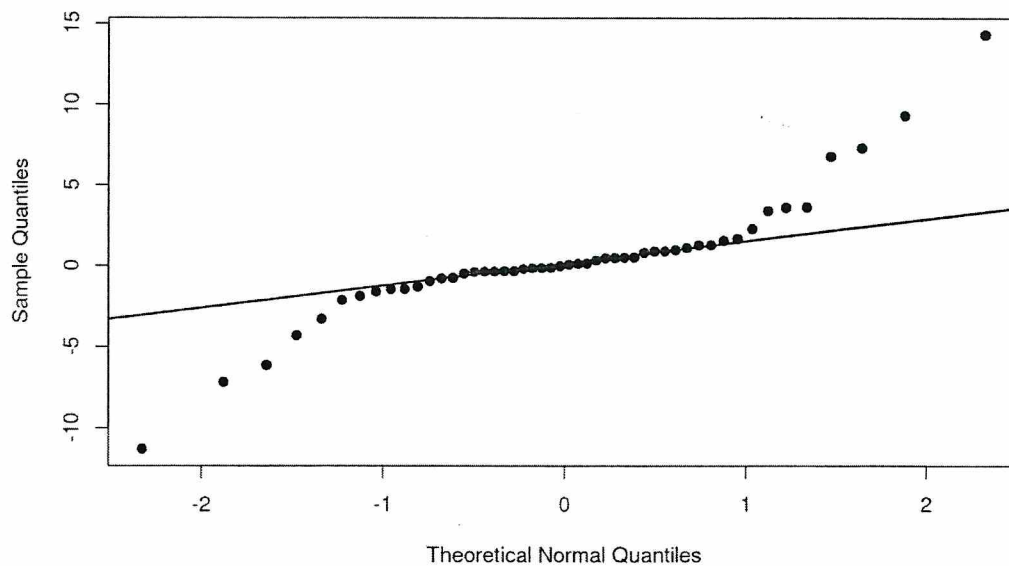
our choice function satisfy ~~the~~ is

$$\lim_{\Delta u \rightarrow 0} \frac{g(u + \Delta u) - g(u)}{\Delta u} = g'(u)$$

$$\Rightarrow \int \frac{g'(u)}{g(u)} du = \int \frac{\frac{C}{u}}{\frac{C}{-u}} du = \int 1 du = u + C$$

Continuing the solution $\rightarrow$ at last

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



Since at extreme value the points are getting spread out in a direction means This high probability of extreme points compare to normal distribution

So ~~the~~ Heavy tail distributions (like t distribution) will characterize it more accurately

02

Q3

$$g(u) = \frac{c}{u} \quad u > 0$$

$$\rightarrow g'(u) = -\frac{c}{u^2} \quad u > 0$$

$$g'(u) = -\frac{c}{u^2} \quad u < 0$$

$$\rightarrow g(u) = c \ln(-u) + K_2 \quad u < 0$$

$$\Rightarrow g(u) =$$

Q3

$$g(u) = \frac{c}{u} \quad u \neq 0$$

$$\Rightarrow g(u) = c \ln|u| + K \quad \forall K \neq 0$$

$g(x) = c \ln|x| + K$ can be chosen for any c, K
 $\forall x \neq 0$ $g(\bar{x}_n)$'s asymptotic variance will be constant

Need to give one such function

05

$$g(x) = \ln|x| \quad x \neq 0$$

Q1

$$R^2 = \frac{(n-1) \sum (\hat{b}_i - \bar{b})^2 / (n-1)}{(n-1) S_{yy}}$$

Sample $\bar{b} = \frac{\sum \hat{b}_i}{n}$
 $\Rightarrow \sum e_i = 0$

Sample variance of $\hat{b}_i$

~~$R^2 = (1-p) \times \text{Sample Variance}(\hat{\beta})$~~

~~$(n-1) S_{yy}$~~

$R^2 = \frac{(n-1) \text{Sample Variance}(\beta^T A_i \beta)}{(n-1) S_{yy}}$

Sample $A_i \beta = \sum \hat{b}_i$
 A_i is i^{th} row of A
 $\hookrightarrow p \times 1$

~~Sample Variance~~

~~$(n-1) A_i^T$~~

~~$= \frac{(n-1) \beta^T S_{xx} \beta}{(n-1) S_{yy}}$~~

Q1

$$R^2 = \frac{SS_{\text{reg}}}{S_{TT}} = \frac{(n-1) \sum (\hat{y}_i - \bar{y})^2 / (n-1)}{(n-1) \sum (y_i - \bar{y})^2 / (n-1)} = \frac{\text{Sample Variance}(\beta^T X_i)}{S_{yy}}$$

$\sum (y_i - \bar{y})^2 / (n-1) \rightarrow S_{yy}$

Sample Variance
 $\beta^T X_1, \beta^T X_2, \dots, \beta^T X_n$
 where X_i is i^{th} row of X

$$R^2 = \frac{\beta^T S_{xx} \beta}{S_{yy}}$$

$$\beta = (X^T X)^{-1} X^T y$$

$$R^2 = S_{yy}^{-1/2} \beta^T S_{xx} \beta S_{yy}^{-1/2}$$

$$\underline{\underline{\beta}} = (X^T X)^{-1} =$$

if $\bar{x}$ observation all mean centered

$$\text{i.e. } X = \begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \dots & x_{1p} - \bar{x}_p \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \dots & x_{2p} - \bar{x}_p \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \bar{x}_1 & x_{n2} - \bar{x}_2 & \dots & x_{np} - \bar{x}_p \end{bmatrix} \quad Y = \begin{bmatrix} y_1 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{bmatrix}$$

$$\bar{x}_i = \frac{\sum_{j=1}^n x_{ij}}{n} \quad \bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$X^T X = S_{XX} (n-1)$$

$$X^T Y = S_{YX} (n-1)$$

$$R^2 = \frac{S_{YX}^2}{S_{YY}} \quad \hat{\beta} = (X^T X)^{-1} X^T Y = (S_{XX} (n-1))^{-1} S_{YX} (n-1) = (S_{XX})^{-1} S_{YX}$$

$$R^2 = S_{YY}^{-1/2} \beta^T S_{XX} \beta S_{YY}^{-1/2}$$

$$= S_{YY}^{-1/2} S_{YX}^T (S_{XX})^{-1} S_{XX} (S_{XX})^{-1} S_{YX} S_{YY}^{-1/2} \quad (\cancel{S_{XX}^T} \Rightarrow S_{XX})$$

$$R^2 = S_{YY}^{-1/2} S_{YX}^T (S_{XX})^{-1} S_{YX} S_{YY}^{-1/2} \rightarrow \textcircled{1} \quad (S_{XX}^T = S_{XX})$$

Hence $R^2 = r^2$ using $\textcircled{1}$ and $\textcircled{2}$
On page $\textcircled{2}$

06